Neural Whittle Index Networks (NeurWIN)

Bypassing Analytical Proofs in RMAB

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Table of Contents

1. Theoretical Foundations
2. Decoupling via Subsidy
3. Indexability and Whittle Indices
4. The NeurWIN Framework
5. Experiments
6. Beyond Independence: Coupled RMABs
7. Literature: Networked RMABs

Table of Contents

1 Theoretical Foundations

2 Decoupling via Subsidy

3 Indexability and Whittle Indices

4 The NeurWIN Framework

5 Experiments

6 Beyond Independence: Coupled RMABs

7 Literature: Networked RMABs

The Exponential Explosion (S^N)

- **The Global Problem:** N restless arms, each with state $s_i \in \mathcal{S}$.
- **Coupling Constraint:** Select exactly $M < N$ arms to activate.
- **Complexity:** Tracking the joint state results in a space of size $|\mathcal{S}|^N$.
- **Example:** If $S = 10, N = 50 \implies 10^{50}$ states. Computationally intractable for traditional MDP solvers.

Table of Contents

1. Theoretical Foundations
2. Decoupling via Subsidy
3. Indexability and Whittle Indices
4. The NeurWIN Framework
5. Experiments
6. Beyond Independence: Coupled RMABs
7. Literature: Networked RMABs

Removing Coupling via Subsidy λ

Decouple the global problem into N independent problems using a subsidy λ for being passive.

- **Modified Reward:** Receive $r(s, a)$ if active $(a = 1)$, or a guaranteed subsidy λ if passive $(a = 0)$.
- **Independence:** Each arm now solves its own individual MDP:

$$\max \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t (r(s_t, a_t) + \lambda(1 - a_t))\right]$$

Table of Contents

1 Theoretical Foundations

2 Decoupling via Subsidy

3 Indexability and Whittle Indices

4 The NeurWIN Framework

5 Experiments

6 Beyond Independence: Coupled RMABs

7 Literature: Networked RMABs

Passive Sets and Indexability

- Let $\mathcal{P}(\lambda)$ be the **Passive Set**: states where it is optimal to be passive given subsidy λ.
- As the "bribe" λ increases, staying passive should logically become more attractive.

Definition (Indexability)

An arm is **indexable** if the passive set $\mathcal{P}(\lambda)$ increases monotonically from $\emptyset$ to $\mathcal{S}$ as the subsidy λ increases.

The Whittle Index $W(s)$

Definition: Whittle Index

The Whittle Index $W(s)$ is the infimum value of the subsidy λ for which it is optimal to be passive in state s:

$$W(s) = \inf\{\lambda : s \in \mathcal{P}(\lambda)\}$$

- **Policy:** Rank arms by $W(s)$ and activate the top M.
- **Benefit:** Complexity drops from $O(S^N)$ to $O(S \times N)$.

The Core Guarantee: Theorem 2

Theorem 2 provides the mathematical "bridge" that allows us to trust a Neural Network for Whittle Index estimation.

The Guarantee

If a policy network $f_\theta(s)$ achieves near-optimal average reward in the simulator, it is **mathematically guaranteed** that $f_\theta(s)$ is an accurate approximation of the true Whittle Index $W(s)$.

- **Certification:** High rewards act as a "certificate" of accuracy.
- **Convergence:** It proves that the "indifference point" found by the Sigmoid policy is the true mathematical root.
- **Trust:** We don't need to see the internal weights; we only need to see the performance.

Table of Contents

1. Theoretical Foundations
2. Decoupling via Subsidy
3. Indexability and Whittle Indices
4. The NeurWIN Framework
5. Experiments
6. Beyond Independence: Coupled RMABs
7. Literature: Networked RMABs

Bypassing the Analytical Proof

- **The Bottleneck:** Traditionally, you must prove indexability manually—often impossible for complex systems.
- **NeurWIN Strategy:** Assume the problem is **Strongly Indexable** (Advantage $D_s(\lambda)$ is strictly decreasing).
- **Discovery via Theorem 2:** High reward empirically "proves" indexability where analytical math fails.

Algorithm 1: NeurWIN Training

Input: Parameters θ, discount factor $\beta \in (0,1)$, learning rate L, sigmoid m
for each mini-batch b **do**
  Randomly choose s_0 and s_1, and set $\lambda \leftarrow f_\theta(s_0)$;
  for each episode e in the mini-batch **do**
    Set the arm to state s_1, and set $h_e \leftarrow 0$;
    for each round t in the episode **do**
      Choose $a[t] = 1$ w.p. $\sigma_m(f_\theta(s[t]) - \lambda)$, and $a[t] = 0$ otherwise;
      if $a[t] = 1$ **then** $h_e \leftarrow h_e + \nabla_\theta \ln(\sigma_m(f_\theta(s[t]) - \lambda))$;
      else $h_e \leftarrow h_e + \nabla_\theta \ln(1 - \sigma_m(f_\theta(s[t]) - \lambda))$;
      end if
    end for
    $G_e \leftarrow$ empirical discounted net reward in episode e;
  end for
  $L_b \leftarrow$ learning rate; $\bar{G}_b \leftarrow$ average of G_e for batch;
  $\theta \leftarrow \theta + L_b \sum_e (G_e - \bar{G}_b) h_e$;
end for

Table of Contents

1 Theoretical Foundations

2 Decoupling via Subsidy

3 Indexability and Whittle Indices

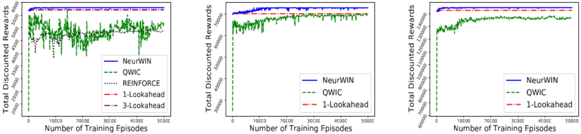
4 The NeurWIN Framework

5 Experiments

6 Beyond Independence: Coupled RMABs

7 Literature: Networked RMABs

- **Concept:** Models time-varying consumer interest; interest drops after a play and recovers over time.

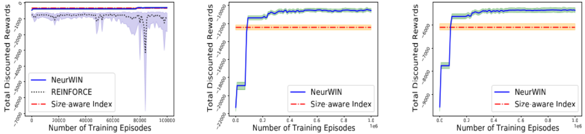


(a) $N = 4$. $M = 1$. (b) $N = 100$. $M = 10$. (c) $N = 100$. $M = 25$.

NeurWIN Training Progress on Recovering Bandits

Exp 5.3: Wireless Scheduling

- **Concept:** Minimize holding costs over fading channels.

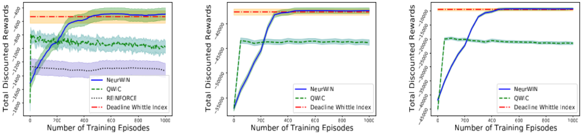


(a) $N = 4.\ M = 1.$ (b) $N = 100.\ M = 10.$ (c) $N = 100.\ M = 25.$

NeurWIN vs Size-aware Index Policy vs REINFORCE

Exp 5.4: Deadline Scheduling

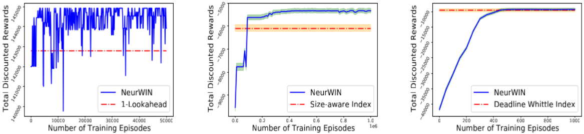
- **Concept:** EV charging station scheduling to maximize reward.



(a) $N = 4.\ M = 1.$ (b) $N = 100.\ M = 10.$ (c) $N = 100.\ M = 25.$

NeurWIN vs Deadline Whittle Index Baseline

- **Setup:** 5% Gaussian noise added to training simulator.



(a) Recovering Bandits. (b) Wireless Scheduling. (c) Deadline Scheduling.

Performance of NeurWIN under Noisy Simulators

Why NeurWIN Performs So Well

The experimental success of NeurWIN across diverse domains is driven by three pillars:

1. **Theorem 2 Security:** It doesn't just "guess"; it is mathematically forced to find the correct index if rewards are high.
2. $S \times N$ **Independence:** By learning per-arm indices instead of joint-state policies, it bypasses the exponential complexity that crushes standard RL.
3. **True Black-Box Capability:** It works directly with simulators. It requires **no** transition matrices, **no** reward formulas, and **no** manual indexability proofs.

Table of Contents

1. Theoretical Foundations
2. Decoupling via Subsidy
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5. Experiments
6. Beyond Independence: Coupled RMABs
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The Limits of Independence

NeurWIN assumes arms are **decoupled** (only linked by a global budget M). In reality, many environments are **Coupled Restless Bandits**.

- **Action-State Coupling (Interference):** Activating Arm A generates noise that directly changes the transition probabilities of Arm B (e.g., 5G small cells).
- **State-State Coupling (Network Effects):** The state of one arm "pulls" the state of another. In epidemic control, a high-infection state in City A increases the risk for City B.

The Challenge: The Whittle Index is "blind" to these relationships because it only looks at one arm at a time.

Beyond Indices: Capturing Arm Relations

Relative Ranking via Deep Architectures

We can extend the network to look past individual states by utilizing deeper, shared layers that process the joint environment.

- **Global Feature Extraction:** Deeper layers process the concatenated state vector $\mathbf{S}$ for cross-arm correlations.
- **Capturing Dependencies:** Learns to model "Synergy" and "Interference."
- **Relative Scoring:** Outputs a **Relative Rank** rather than an absolute index, ensuring the top M arms complement each other.

Table of Contents

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2. Decoupling via Subsidy
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5. Experiments
6. Beyond Independence: Coupled RMABs
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Modern Solutions for Coupled Systems

Recent research bridges the gap between independent Whittle logic and networked dependencies through new input-output strategies.

Paper	Key Input	Mechanism	Selection Strategy
Killian et al. (AAAI '21)	Arm State + Adjacency Matrix	Networked Whittle Index (W_{net})	**Ranking:** Rank by spillover-aware scalar.
coRMAB (NeurIPS '24)	Joint State $\mathbf{S}$ + Constraints $\mathcal{C}$	Deep RL + MILP Optimizer	**Decision:** Outputs a feasible binary vector.
GNB (ICLR '23)	Node Features + Message Passing	Graph Neural Networks (GNN)	**Relative Score:** Contextualized reward estimate.

- **Input Shift:** Moving from single-state s_i to graph-aware features.
- **Output Shift:** Moving from an *absolute* index to a *negotiated* priority or direct combinatorial decision.